

Raritone Summer Internship 2026

Task: Comparison of AR/VR Frameworks for Virtual Try-On Systems

1. Introduction

Virtual Try-On systems allow users to visualize clothes, shoes, jewellery, and accessories on a digital avatar or directly on themselves using Augmented Reality (AR) and Virtual Reality (VR) technologies. Selecting the right framework is critical for achieving realistic rendering, accurate tracking, cross-platform compatibility, and smooth user experiences.

This report compares four major frameworks used in AR/VR development:

1. Unity AR Foundation
2. ARCore
3. ARKit
4. Three.js

The comparison focuses on platform support, performance, development complexity, AI integration, and suitability for Raritone's Virtual Try-On platform.

2. Unity AR Foundation

Overview

Unity AR Foundation is a cross-platform framework developed by Unity Technologies. It provides a unified API for developing AR applications that work on both Android and iOS devices.

Key Features

- Single codebase for Android and iOS
- Plane detection
- Face tracking
- Object tracking
- Environment understanding

- Integration with Unity's 3D engine
- Support for ARCore and ARKit

Advantages

- Cross-platform development
- Strong 3D rendering capabilities
- Large developer community
- Easy integration with AI/ML models
- Supports advanced AR experiences

Limitations

- Larger application size
- Higher hardware requirements
- Steeper learning curve for beginners

Best Use Cases

- Virtual clothing try-on
- Real-time avatar visualization
- Interactive AR shopping applications

3. ARCore

Overview

ARCore is Google's augmented reality platform designed specifically for Android devices.

Key Features

- Motion tracking
- Environmental understanding
- Light estimation
- Depth sensing
- Image recognition

Advantages

- Excellent Android support
- Accurate tracking
- Optimized performance

- Strong integration with Android ecosystem

Limitations

- Android only
- Limited support outside Google's ecosystem

Best Use Cases

- Android-based AR applications
- Mobile virtual try-on systems
- Location-based AR experiences

4. ARKit

Overview

ARKit is Apple's augmented reality framework designed exclusively for iOS devices.

Key Features

- Face tracking
- Body tracking
- Motion tracking
- Scene understanding
- LiDAR support on compatible devices

Advantages

- Industry-leading tracking accuracy
- Excellent optimization
- Smooth user experience
- High-quality rendering

Limitations

- Available only on Apple devices
- Requires iOS ecosystem

Best Use Cases

- Premium AR experiences
- iPhone and iPad applications
- High-accuracy virtual try-on systems

5. Three.js

Overview

Three.js is a JavaScript library used for creating 3D graphics and AR/VR experiences directly in web browsers.

Key Features

- Browser-based 3D rendering
- WebGL support
- Lightweight deployment
- Easy web integration
- Interactive 3D environments

Advantages

- No application installation required
- Works directly in browsers
- Easy deployment
- Cross-platform accessibility

Limitations

- Lower performance compared to native frameworks
- Limited advanced AR features
- Browser hardware restrictions

Best Use Cases

- Web-based virtual try-on
- Product visualization
- Lightweight AR applications

6. Framework Comparison

Feature	Unity AR Foundation	ARCore	ARKit	Three.js
Platform Support	Android & iOS	Android	iOS	Web Browser
AR Support	Excellent	Excellent	Excellent	Moderate

Feature	Unity AR Foundation	ARCore	ARKit	Three.js
VR Support	Yes	Limited	Limited	Yes
Performance	High	High	Very High	Medium
Ease of Development	Medium	Medium	Medium	Easy
AI Integration	Excellent	Good	Good	Moderate
Cross Platform	Yes	No	No	Yes
Community Support	Very Large	Large	Large	Large
Real-Time 3D Rendering	Excellent	Good	Excellent	Good
Best Use Case	Full AR Platform	Android AR	iOS AR	Web AR

7. Pros and Cons Summary

Framework	Pros	Cons
Unity AR Foundation	Cross-platform, powerful rendering, AI friendly	Larger app size
ARCore	Strong Android support, accurate tracking	Android only
ARKit	Best tracking quality, optimized performance	iOS only
Three.js	Browser-based, easy deployment	Lower performance than native AR

8. Analysis for Raritone

The Raritone Virtual Try-On platform requires:

- Real-time AR visualization
- Accurate body tracking
- AI model integration
- 3D avatar support
- Cross-platform deployment

- Scalability for future features

After evaluating all frameworks, Unity AR Foundation provides the most balanced solution because it allows developers to create a single application that supports both Android and iOS devices while leveraging ARCore and ARKit internally.

Three.js is also a strong option for browser-based virtual try-on experiences where users do not want to install an application.

9. Final Recommendation

Recommended Framework

Primary Choice

Unity AR Foundation

Reasons:

- Cross-platform support
- Advanced AR capabilities
- Strong AI integration
- Excellent 3D rendering
- Suitable for avatar-based virtual try-on

Android Support

ARCore

iOS Support

ARKit

Web-Based Solution

Three.js

10. Conclusion

The comparison demonstrates that each framework serves a specific purpose. ARCore and ARKit provide excellent native AR experiences for Android and iOS respectively, while Three.js enables browser-based AR solutions. However, for a complete Virtual Try-On platform such as Raritone, Unity AR Foundation is the most suitable framework due to its cross-platform capabilities, powerful rendering engine, AI compatibility, and support for scalable AR applications.

Therefore, Unity AR Foundation is recommended as the primary framework for future development of the Raritone Virtual Try-On system.